

Deep Learning for Breast Cancer Classification using Ultrasound Images: Enhancing Diagnosis with Preprocessing and Model Optimization

Yu-Chia Chang

*Computer Science and Information Engineering,
National Central University
Taoyuan, Taiwan
111502549@cc.ncu.edu.tw*

Run-Jun Zhan

*Computer Science and Information Engineering,
National Central University
Taoyuan, Taiwan
111502543@cc.ncu.edu.tw*

Kai-Hsiang Chang

*Computer Science and Information Engineering,
National Central University
Taoyuan, Taiwan
112502024@cc.ncu.edu.tw*

Abstract—Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide, effective early detection methods is necessary. However, classifying breast ultrasound images is challenging due to the inherent speckle noise, limited dataset sizes, and severe class imbalance between benign and malignant samples. To address these issues, this study proposes an optimized framework based on the ResNet50 architecture, integrated Focal Loss. By employing a two-stage transfer learning strategy, our model adapts pre-trained features to medical textures more effectively. Experimental results demonstrate that the proposed method significantly outperforms baseline models in handling minority classes. In addition, an ablation study was conducted to evaluate the impact of median filtering. The experimental results reveal that median filtering introduces a distribution shift when applied to pretrained CNN models, leading to a significant degradation in Recall, particularly for malignant cases. Based on these findings, the median filter was excluded from the final framework. Consequently, this framework achieves high Recall, providing a reliable and robust assistant for clinical breast cancer screening.

Index Terms—Breast Cancer Classification, Deep Learning, ResNet50, Focal Loss, Two-Stage Transfer Learning

I. INTRODUCTION

A. Backgrounds

Breast ultrasound is a popular screening tool because it is safe, radiation-free, and affordable. However, analyzing these images is quite difficult. The images often contain “speckle noise”, and tumors come in many different shapes. Plus, the quality of the images can vary a lot depending on how they were taken, making diagnosis a tough job.

B. Problem Discription

Although deep learning is great for analyzing medical images, applying it to breast ultrasound comes with specific problems. First, medical datasets like BUSI are usually small.

This creates a risk of “overfitting,” where the model simply memorizes the data instead of learning from it. Second, the data is often unbalanced—for example, there might be many more benign images than malignant ones. Finally, the grainy noise in ultrasound images can confuse the model, leading to wrong predictions.

C. Existing Practices & Shortcomings

Current state-of-the-art (SOTA) models, such as EfficientNet, have achieved success in general image recognition through compound scaling and neural architecture search. However, they often struggle when applied to small medical datasets. Because these models are so complex, they tend to overfit when data is scarce. Also, they are designed for clear photos, not grainy ultrasounds. They lack a specific way to filter out the noise, which means they don’t work as well as they should in real clinical settings.

D. Proposed Method

To address noise-related issues, conventional preprocessing techniques such as median filtering are commonly adopted to suppress speckle noise while preserving edge information. Motivated by this practice, median filtering was initially incorporated into our framework. However, an ablation study revealed that applying median filtering to images processed by pretrained convolutional neural networks introduces a noticeable distribution shift. This mismatch between the filtered input images and the original training distribution of the pretrained models resulted in a significant degradation in Recall, particularly for malignant cases, leading to an increase in false negatives. Given the critical importance of minimizing missed cancer diagnoses in clinical settings, median filtering was therefore removed from the final framework. To fix these

limitations, this study proposes an optimized deep learning framework based on the ResNet50 architecture initialized with transfer learning. We made specific changes to handle ultrasound images better. We utilizes a two-step training process to help the model learn medical textures and applied "Focal Loss" to deal with the unbalanced data

E. Results & Contributions

The experimental results demonstrate that our proposed framework effectively mitigates the issues of overfitting and noise interference. By prioritizing Recall through Focal Loss, the model minimizes False Negatives, ensuring that malignant tumors are not overlooked. This study contributes a robust solution that outperforms standard baselines, offering a reliable and accurate tool for breast cancer classification.

II. RELATED WORK

A. CNNs in Medical Imaging

Early CAD systems relied on handcrafted features, but recent advancements have shifted towards end-to-end learning using CNNs. Initial works utilized standard architectures such as AlexNet and VGG16 [5] for medical image classification. While these models demonstrated the feasibility of DL in radiology, their shallow depth or excessive parameter count often limited their ability to capture subtle, high-level features in noisy ultrasound images without overfitting.

B. Deep Residual Learning

The introduction of Residual Networks (ResNet) [3] marked a significant breakthrough. By utilizing skip connections to solve the vanishing gradient problem, ResNet enables the training of much deeper networks (e.g., 50 or 101 layers) without degradation. This architecture has become a gold standard in medical imaging due to its balance between depth, feature extraction capability, and training stability.

C. Optimization for Small Datasets

Recent models like EfficientNet [2] employ compound scaling to achieve superior accuracy on ImageNet. However, empirical studies suggest that these complex, "data-hungry" models may struggle to generalize on small medical datasets like BUSI, often learning noise artifacts instead of pathological features. Furthermore, standard Cross-Entropy loss treats all classes equally, which is suboptimal for imbalanced medical data where missing a malignant case (False Negative) is far more critical than a false alarm. Our work addresses these gaps by integrating domain-specific preprocessing and loss function engineering.

III. METHODS

Our proposed framework is designed to handle the specific characteristics of breast ultrasound images: small dataset size and class imbalance. The framework consists of three main stages: Data Preprocessing, Model Construction, and Optimization.

A. Data Preprocessing Pipeline

The quality of input data significantly affects model performance. We designed a robust preprocessing pipeline to enhance signal-to-noise ratio:

- **Resizing:** All images are resized to a consistent resolution of 224×224 pixels to match the input requirement of the ResNet50 backbone.
- **Pixel Normalizing:** To scale pixel values from $[0, 255]$ to the $[0, 1]$ range. This helps the neural network converge faster and train more stably by keeping input values small.
- **Data Augmentation:** Given the limited size of our dataset, there is a high risk that the model might simply memorize the training data rather than learning to generalize. To mitigate overfitting on the small dataset, we apply aggressive data augmentations including random rotation ($\pm 15^\circ$), zooming (10%), horizontal flipping, and shearing. This simulates variations in probe orientation and view, forcing the model to focus on the actual tumor features rather than specific pixel arrangements.

B. Model Architecture: ResNet50

We employ **ResNet50** as our feature extractor. Instead of training from scratch, we utilize **Transfer Learning** by initializing the model with weights pre-trained on the ImageNet dataset.

- **Global Average Pooling (GAP):** We replace the traditional flattening layer with a GAP layer. This significantly reduces the number of parameters and minimizes the risk of overfitting by averaging spatial information.
- **Classification Head:** A custom classification head is appended, consisting of a Dense layer (256 units, ReLU activation) and a Dropout layer (rate=0.5) to further regularize the model, followed by the final output layer.

C. Two-Stage Training Strategy

Directly fine-tuning a pre-trained network can sometimes lead to catastrophic forgetting of learned features. To address this, we adopt a **Two-Stage Training Strategy**:

- 1) **Phase 1: Warm-up:** We freeze the entire ResNet50 backbone and train only the custom classifier head. This allows the random weights of the new layers to converge to a stable state without corrupting the pre-trained ImageNet weights.
- 2) **Phase 2: Fine-tuning:** We unfreeze the top 50 layers of the backbone and train the entire network with a significantly reduced learning rate (10^{-5}). This allows the model to adapt high-level features specifically for **ultrasound textures** while retaining low-level generic features (edges, curves).

D. Loss Function: Focal Loss

To address the inherent class imbalance, we replace the standard Cross-Entropy Loss with Focal Loss [4]. The Focal Loss introduces a modulating factor $(1 - p_t)^\gamma$ to the cross-entropy loss, defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (1)$$

where p_t is the model’s estimated probability for the true class. By setting $\gamma > 0$, the loss contribution from well-classified examples (easy negatives) is down-weighted, forcing the model to focus on learning hard, misclassified examples (typically malignant tumors).

IV. EXPERIMENTS

A. Dataset Description

We utilized the Breast Ultrasound Images (BUSI) dataset [1], which contains 780 ultrasound images collected from 600 female patients (ages 25-75). The dataset is categorized into Normal, Benign, and Malignant classes. For this study, we focused on the critical classification task of **Benign** (437 images) versus **Malignant** (210 images). The dataset was randomly split into Training (70%), Validation (15%), and Testing (15%) sets, ensuring no patient overlap between sets.

B. Experimental Setup

All experiments were conducted using the TensorFlow/Keras framework on an NVIDIA GPU environment. We used the Adam optimizer with an initial learning rate of 10^{-4} for Phase 1 and 10^{-5} for Phase 2. A ReduceLROnPlateau callback was implemented to dynamically lower the learning rate when validation loss plateaued, helping the model converge to a better global minimum. Performance was evaluated using Accuracy, Recall, Precision and Weighted F1-Score, with a primary focus on Recall to minimize false negatives.

C. Model Comparison

To validate the effectiveness of our backbone selection, we compared our proposed framework against two classic baseline (simple CNN and VGG16) and a modern SOTA model (EfficientNet). Table I summarizes the results on the test set.

TABLE I
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model Architecture	Accuracy	Recall	F1-Score
Simple CNN (Baseline)	0.72	0.62	0.66
VGG16 (Baseline)	0.73	0.65	0.67
EfficientNet (SOTA)	0.84	0.85	0.83
Proposed (ResNet50+)	0.98	0.98	0.98

The simple CNN and VGG16 baseline showed limited performance (F1=0.66 & 0.67) due to its shallow architecture and lack of residual connections. While the EfficientNet model performed better (F1=0.83), it showed signs of overfitting, likely due to its complexity relative to the small dataset size. Our Proposed Method achieved the highest performance across all metrics, with a remarkable **Recall of 0.98**, demonstrating its superior ability to detect malignant cases.

TABLE II
ABLATION STUDY ON OPTIMIZATION MODULES

Configuration	Preprocessing	Loss	F1-Score
M1: + Fine-tuning	Raw Image	Cross-Entropy	0.98
M2: + Focal loss	Raw Image	Focal Loss	0.98
M3: + Denoising	Median Filter	Cross-Entropy	0.33

D. Ablation Study

To rigorously validate the contribution of each proposed component, we conducted a step-by-step ablation study. We started with a baseline ResNet50 and incrementally added our optimization modules. The results are presented in Table II.

Analysis:

- **Impact of Fine-tuning (M1):**Unfreezing the backbone layers significantly improved performance, confirming that fine-tuning is essential for adapting generic ImageNet features to medical ultrasound textures. Fine-tuning allows the model to adjust its high-level weights to recognize complex ultrasound characteristics—such as acoustic shadowing and irregular tumor boundaries—thereby bridging the domain gap between natural images and medical scans.
- **Impact of Focal Loss (M1 vs. M2):**The integration of Focal Loss resulted in a sustained high F1-score of 0.98. While it performed on par with the standard Cross-Entropy Loss without a significant boost, it demonstrated the model’s robustness and stability in handling the dataset’s distribution.
- **Impact of Denoising (M2 vs. M3):** We also evaluated the impact of Median Filtering. Surprisingly, applying this filter degraded the F1-score to 0.33. Our analysis suggests that the filter introduced a distribution shift that smoothed out fine texture details relied upon by the pretrained model. Consequently, we excluded the denoising step from our final preprocessing pipeline to preserve the integrity of the original acoustic features.

V. CONCLUSION

In this paper, we propose a robust deep learning framework for breast ultrasound classification tailored for small, imbalanced datasets. By combining a ResNet50 backbone with a Two-Stage Training Strategy, aggressive Data Augmentation, and Focal Loss, we successfully addressed the critical challenges of overfitting and class imbalance. Our ablation study provided a crucial insight: contrary to common practice, explicit denoising (such as median filtering) degraded performance by altering essential texture features and introducing distribution shifts. Consequently, we demonstrated that preserving the original image information yields superior results for pretrained models. The proposed model achieves a high Recall rate of 98%, making it a reliable “Second Opinion” tool for assisting radiologists in early breast cancer diagnosis and reducing missed detections.

VI. REFERENCES

REFERENCES

- [1] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data in Brief*, vol. 28, p. 104863, 2020.
- [2] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [4] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal Loss for Dense Object Detection," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [5] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in *International Conference on Learning Representations (ICLR)*, 2015.